

MANUFACTURING PROCESSES-I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$
 - i. The hot working of metals is carried out -
 - a) At the recrystallization temperature
 - b) Below the recrystallization temperature
 - c) Above the recrystallization temperature
 - d) At any temperature
 - ii. Mechanical property desired for successful forging is - (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness.
 - iii. In a four high rolling mill, there are four rolls out of which -
 - a) One is working roll and three are backing up rolls
 - b) Two are working rolls and two are backing up rolls
 - c) Three are working rolls and one is backing up roll
 - d) All of the four are working rolls
 - iv. Seamless tubes in mass production are manufactured by - (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - v. Blanking and punching operations can be performed simultaneously on -
 - (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - vi. Lathe bed is made of a) Mild Steel b) High Carbon steel c) Cast Iron d) High Speed Steel
 - vii. The angle between face and flank of a single point cutting tool is known as a) Rake Angle b) Clearance Angle c) Lip Angle d) Nose Angle
 - viii. Which is an example of orthogonal cutting? (a) Milling, (b) Turning (c) Drilling, (d) Planning.
 - ix. Which is related with offsetting the tail stock? a) Turning b) Facing c) Taper turning d) Knurling.
 - x. The allowance given for easy removal of pattern is a) Shrinkage allowance b) Wrapping Allowance c) Machining Allowance d) Draft Allowance
 - xi. Gate is provided in moulds to - (a) Feed the casting at a constant rate, (b) Give passage to gases, (c) Compensate for shrinkage, (d) Avoid cavities.
 - xii. Metal and alloys of which following metal is used for cold chamber die casting-(a) tin, (b) lead, (c) aluminium, (d) iron
 - xiii. The flame used for welding brass is - a) Oxidising b) Neutral c) Carburising d) Natural Flame
 - xiv. The welding arc is not visible to worker in a) MMAW b) GTAW c) GMAW d) SAW
 - xv. Spot welding is one of the type of a) Arc Welding b) Resistance Welding c) Pressure Welding d) Solid-Liquid State welding

GROUP-B

Answer any Five (05) questions.

2. i) What are the advantages of hot working over cold working process?
ii) With suitable diagram explain the open die forging process. 4+4
 3. i) What do you mean by rolling? Discuss 2 Hi and 3 Hi roll stand arrangement with sketches.
ii) Name two forging defects and state the reason for arising these defects. (1+5)+2
 4. i) Discuss forward extrusion and backward extrusion with neat sketches.
ii) Describe sheet metal operations (any two) with sketch. a) Blanking, b) Punching c) Drawing. 4+4
 5. i) State the desirable properties of cutting tool materials.
ii) Explain the different types of chips formed during machining with sketches. 2+6
 6. i) Differentiate between orthogonal cutting and oblique cutting.
ii) Define Cutting speed, feed and depth of cut in Machine tool.
iii) State the functions of cutting fluid. 3+3+2
 7. i) Explain the nomenclature of a single point cutting tool with a neat sketch.
ii) Metric thread of 0.8 mm pitch is to be cut on a lathe. Pitch of the lead screw is 1.5 mm. Find the change gear ratio. 6+2
 8. i) Draw and label the elements of a gating system and briefly explain the function of each.
ii) Explain the causes and remedies of three casting defects with suitable sketches. 5+3
 9. i) Describe the different types of pattern allowances in foundry work.
ii) State the difference between MIG welding and TIG welding. 5+3
 10. i) Describe the characteristics and applications of different gas welding flames with neat sketches.
ii) Distinguish between brazing and soldering. 6+2
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MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.
 This answer is to be made in separate loose script(s) provided for the purpose.
 Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and
 fresh answer scripts for answering the remaining part of the question will be provided.
 On early submission of answer scripts of Question No.1,
 a student will get the remaining script earlier.

Answer any five questions from the rest.

1. A. Fill in the blanks (any ten): 1x10
 - i) The ratio of _____ and acetylene in neutral flame is 1:1.
 - ii) Electrode material for _____ welding is Tungsten.
 - iii) _____ flame is used for welding brass.
 - iv) Axis of the casting and axis of rotation is different in _____ casting process.
 - v) In _____ die-casting method continuous heating of metal is not necessary.
 - vi) _____ allowance is given to compensate effect after solidification.
 - vii) _____ allowance is given for easy removal of pattern.
 - viii) Master pattern is used for making wax pattern for _____ casting method.
 - ix) In _____ welding process arc is not visible to welder.
 - x) Seam welding may be considered as continuous _____ welding.
 - xi) _____ welding is a modification of spot welding.
 - xii) Core box is used to make _____.
 - xiii) Lathe is specified by _____ of the job over carriage.
 - xiv) A small depression in the parent metal during arc welding is called _____.
- B. Choose the correct answer (any ten): 1x10
 - i) Solid / split / sweep pattern is made in a single piece.
 - ii) Spot / TIG / Gas welding is an example of resistance welding.
 - iii) Spot / Seam / TIG welding is used in automobile industry.
 - iv) Electrode is not consumed during TIG / MIG / spot welding.
 - v) Projection / Gas / TIG is an example of resistance welding.
 - vi) Spelter is used during soldering / brazing / TIG.
 - vii) Spot welding electrode is made off copper / aluminium / iron.
 - viii) Brazing is similar to soldering / Gas welding / oxy-cutting.
 - ix) Skeleton pattern is used when pattern is big / small.
 - x) Casting is a forming / deforming process.
 - xi) Sanding is a surface finishing / machining process.
 - xii) Brazing is a joining process / removing process.
 - xiii) The filler material used during soldering is called solder / spelter.
 - xiv) Segmental pattern is used for circular / square work.
2. Describe hot chamber and cold chamber die-casting method.
3. Write short notes on – (a) Sweep pattern, (b) Loose-piece pattern. 5+5
4. Describe submerged arc welding process. 5+5

5. What is core print? What is core? Write the advantages of wood as pattern making material. 5+5
 6. How a Lathe machine can be specified? Write the functions of Lathe machine parts? 5+5
 7. Write the difference between up milling and down milling. Write the specifications of drilling machine. 5+5
 8. Write shorts on – (a) Facing, (b) Parting-off, (c) Knurling. 3+3+4
 9. Describe different types of milling cutter. 10
 10. Write short notes on different types of chips. 10
 11. Write the functions of different parts of a radial drilling machine. 10
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December 2018

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70***Answer to Question No.1 is compulsory and to be answered first.****This answer is to be made in separate loose script(s) provided for the purpose.****Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.****On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.****Answer any five questions from the rest.**

- I. Answer the following questions (any twenty): 1x20
- i) Mechanical property desired for successful forging is – (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness
 - ii) Seamless tubes in mass production are manufactured by – (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - iii) In drop forging, forging is done by dropping – (a) The work piece at high velocity, (b) The hammer at high velocity, (c) The die with hammer with high velocity, (d) A weight on hammer to produce the requisite impact.
 - iv) Blanking and punching operations can be performed simultaneously on – (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - v) Fluxes are used while welding – (a) To increase the rate of welding, (b) To clean the joint, (c) To prevent oxidation of metal during welding, (d) All the above.
 - vi) Oxy-acetylene flame is used for welding of – (a) Monel, (b) Nickel, (c) Steel, (d) Bronze.
 - vii) The hidden defect of a weld can be detected by – (a) Tensile test, (b) Bend test, (c) Radiography, (d) All.
 - viii) Commonly used gases in TIG welding are – (a) Hydrogen & Oxygen, (b) Hydrogen & Argon, (c) Hydrogen & Helium, (d) Helium & Argon.
 - ix) Consumable electrode is used in the following process – (a) TIG welding, (b) Gas welding, (c) Arc welding, (d) Resistance welding.
 - x) Above _____ temperature, working of metals is named hot working.
 - xi) Drawing is done to _____ the diameter of wire.
 - xii) In 4-high rolling mill, bigger rolls are known as _____ rolls.
 - xiii) Accurate taper and step tapers can be machined with _____ attachment.
 - xiv) Finishing and sizing a hole is called _____.
 - xv) A twist drill has _____ flutes.

- xvi) Pattern used in precision investment casting normally made of _____.
- xvii) In resistance welding _____ is generated by the resistance offered to the flow of current.
- xviii) MIG welding uses _____ electrode.
- xix) In neutral flame in Oxyacetylene gas welding, ratio of acetylene to oxygen is _____.
- xx) Spot welding is one of the _____ welding processes.
- xxi) The revolving cutter of a milling machine is held on a _____.
- xxii) In universal milling machine table can have the _____ arrangement about a vertical axis.
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2. a) What is Pattern? Name different types of pattern. Sketch any three of them and explain.
 b) What are the pattern making allowances? Explain any one of them. 7+3
3. a) Discuss with sketches hot rolling and cold rolling.
 b) Discuss with neat sketches direct extrusion and indirect extrusion. 5+5
4. a) How do you specify a Lathe machine?
 b) Write short notes on orthogonal cutting and Oblique cutting.
 c) What are the different operations that can be done on a lathe machine? 3+5+2
5. a) How do you specify a Drilling machine? Draw a pillar type Drilling machine & mention basic parts.
 b) Write short notes on Up Milling & down Milling. 7+3
6. a) What is core? What is the purpose of core?
 b) Explain the defects in casting. How can those defects be prevented? 5+5
7. a) Explain Arc welding process showing all salient points on it.
 b) State various welding defects & explain the remedies. 5+5
8. a) Explain spot welding and show its simple set up.
 b) Name the different flames and show the different zones in each of them. 5+5
9. Write short notes on the followings (any four): 2½x4
 (a) Knurling, (b) Reaming, (c) Brazing, (d) Soldering, (e) Core Prints, (f) Centrifugal Casting, (g) Die Casting
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MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.

**This answer is to be made in separate loose script(s) provided for the purpose.
Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.**

**On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.**

Answer any five questions from Group-A, B & C, taking at least one from each group.

1. Answer the following questions (any twenty): 1x20
 - i) In Oxyacetylene gas welding with neutral flame, the ratio of acetylene to oxygen is – (a) 1:2, (b) 1:1, (c) 2:1, (d) 1:3.
 - ii) Non-consumable electrode is used in the – (a) MIG welding, (b) TIG Welding, (c) Arc welding, (d) Gas welding.
 - iii) Chances of crack propagation are more in – (a) cold working process, (b) hot working process, (c) both (a) & (b), (d) none of these.
 - iv) The gating ratio 2:8:1 for copper in gating system design refers to the ratio of areas of – (a) sprue : runner : ingate, (b) runner: ingate : sprue, (c) runner : sprue : ingate, (d) ingate : runner : sprue.
 - v) The centre lathes receive their power through – (a) headstock, (b) tailstock, (c) both (a) & (b), (d) none of these.
 - vi) In oblique cutting system, the maximum chip thickness – (a) occurs at the middle, (b) may not occur at the middle, (c) depends upon the material of the tool, (d) depends upon the geometry of the tool.
 - vii) Which current is used in Tungsten Inert-Gas (TIG) welding? – (a) Only A.C. can be used as welding current, (b) Only D.C. can be used as welding current, (c) Both A.C. and D.C. can be used as welding current, (d) None of these.
 - viii) Gate is provided in moulds to – (a) Feed the casting at a constant rate, (b) Give passage to gases, (c) Compensate for shrinkage, (d) Avoid cavities.
 - ix) A moving mandrel is used in – (a) wire drawing, (b) tube drawing, (c) metal cutting, (d) forging.
 - x) Scab is a – (a) welding defect, (b) sand casting defect, (c) machining defect, (d) forging defect.
 - xi) Which of the following gas mixtures is not used in Gas Tungsten arc welding (TIG)? – (a) Argon-Helium, (b) Argon-Nitrogen, (c) Argon-Hydrogen, (d) Argon-CO₂.
 - xii) Which of the following is always a cold working operation? – (a) Rolling, (b) Wire drawing, (c) Forging, (d) Extrusion.
 - xiii) The angle between face and flank of a single point cutting tool is known as _____.
 - xiv) In _____ welding process, the arc is not visible to worker.

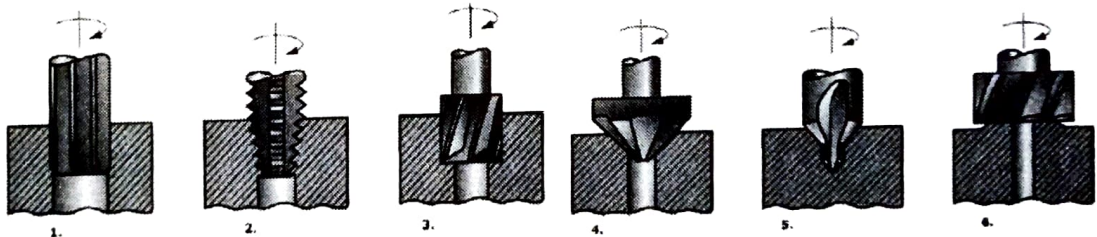
- xv) _____ is a mixture of finely divided aluminum and iron oxides.
- xvi) Cuts and washes in sand casting is caused by _____ of molding sand by the flowing metal.
- xvii) Spot/Seam/TIG welding is used in automobile industry.
- xviii) Hot working of metal in metal forming processes is working _____ recrystallization temperature.
- xix) In MIG welding, _____ current _____ polarity is generally used.
- xx) Single point cutting tool should ideally have _____ rake angle.
- xxi) A lathe bed is generally made of _____.

Group-A

- 2.
 - a) What do you mean by forming? What is recrystallization?
 - b) Distinguish between forward extrusion and backward extrusion with neat sketches.
 - c) What are the advantages of hot working over cold working process? (1+2)+4+3
- 3.
 - a) What do you mean by rolling? Discuss different kinds of roll stand arrangement.
 - b) State three forging defects and state the reason for arising these defects. (1+4)+2+3
- 4.
 - a) Discuss different stages of drop forging.
 - b) What is the difference between punching and blanking? What is meant by stripping force and punching force? 5+5

Group-B

- 5.
 - a) Differentiate between orthogonal cutting and oblique cutting.
 - b) Explain the nomenclature of a single point cutting tool with a neat sketch. 4+6
- 6.
 - a) Differentiate between Capstan and Turret lathe.
 - b) Identify the following machining operations and define them. 4+6



- 7.
 - a) How do you specify a milling machine?
 - b) Differentiate between conventional milling & climb milling.
 - c) Write short notes on any two – (a) Knurling, (b) facing, (c) Parting off. 2+4+4
- 8.
 - a) What are the different types of chips formed during machining? Write short note on Continuous chip with built-up edges.
 - b) Draw the neat sketch of a twisted drill bit and mark various part of it. (1+3)+6

Group-C

- 9.
 - a) Define core print. What is the use of it?
 - b) Write down the advantages and disadvantages of green sand mould.
 - c) Write short notes on – (i) Sweep pattern, (ii) Loose-piece pattern. 2+4+4

10. a) Draw the diagram depicting two-piece sand casting arrangement and show the locations of pouring basin, sprue, gate, runner, riser, mould cavity, parting line, air vent, cope and drag on it.
b) Name three welding defects. State the reason for arising any one of such defects and the remedies to avoid them. 5+(1+2+2)
11. a) What are the typical electrode materials used for Tungsten Inert gas welding?
b) Explain the characteristics of Arc welding machines, which viz., constant current and constant voltage. Mention the applications of each of them.
c) What are the problems encountered with the use of coated electrodes? Explain how these can be taken care of. 2+4+4
12. a) Distinguish between – (i) hot chamber and cold chamber die casting, (ii) Soldering & Brazing.
b) Explain precision investment casting process with neat sketch. (3+3)+4
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DECEMBER 2024

3236244**MANUFACTURING PROCESS - I***Time Allowed: 3 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- (i) The point at which the cutting tool reaches, beyond which it will not function satisfactorily until it is reground, is called as: a) Tool wear, b) Tool failure, c) Tool diffusion, d) None of the above.
- (ii) Which cutting condition affects the cutting temperature predominantly?
 a) Depth of cut, b) cutting speed, c) feed, d) All of the above.
- (iii) In a 4-Hi rolling mill, backup rollers are used to _____
 a) Reduce rolling force, b) Increase stiffness of rolling mill, c) Reduce deflection of working rollers, d) Increase reduction in thickness of job.
- (iv) Shaping of metal by application of compressive force in between two or more dies in order to obtain desired shape is done by- a) Forging, b) Rolling, c) drawing, d) blanking.
- (v) In which of the following forging metal is kept in the lower die?
 a) Open die, b) closed die, c) hold die, d) impression die.



- (vi) The metal forming process in blanking operation is associated with which type of force?
 a) Shear force, b) tensile force, c) compressive force, d) all of the above.
- (vii) The energy requirement in direct extrusion is _____ than indirect extrusion.
 a) More, b) less, c) same, d) no correlation.
- (viii) The surface finish of work piece in hot working is _____ the surface finish in cold working.
 a) Better than, b) worse than, c) same as, d) no correlation.
- (ix) By which material is lathe bed made?
 a) High carbon steel, b) cast iron, c) mild steel, d) copper.
- (x) For which kind of chuck is centering not required?
 a) 4-jaw chuck, b) magnetic chuck, c) 3-jaw chuck, d) mandrel.



3236244

- (xi) Which type of feed is needed in facing operation?
a) Longitudinal, b) cross, c) both longitudinal and cross, d) angular.
- (xii) In neutral flame acetylene and oxygen is in _____ by volume.
a) 1:1 ratio, b) 2:1 ratio, c) stoichiometric ratio, d) ratio.
- (xiii) The ability of the moulding sand to withstand the heat of melt without showing any sign of softening is called as _____.
a) Strength of cohesiveness, b) refractoriness, c) collapsibility, d) flowability.
- (xiv) Which of the following is not a requirement of a good pattern material?
a) It should be light in weight, b) it should have smooth surface, c) it should be easily available, d) it should have low strength to break it and remove easily from mould.
- (xv) AC arc in arc welding is used for welding _____.
a) Hard material, b) highly oxidative material, c) ductile material, d) any material.

GROUP-B

2. (i) Classify manufacturing processes. 3
(ii) What is the main advantage of die forging over impact forging? 2
(iii) Discuss about any two Forging Defects. 3
3. (i) Classify rolling mill as per number of rollers. Draw figures for each of them. 3
(ii) Why smaller roller should be used to decrease contact surface for same deformation? 2
(iii) Write differences between direct and indirect extrusion. 3
- * * * * *
4. (i) Briefly discuss about Different components and their functions of standard die set with a punch. 5
(ii) What are punching and blanking operation? 3
5. (i) Draw three views of a single point cutting tool showing different angles, radius and planes in appropriate views. 3
(ii) Show the specifications of a single point cutting tool having nomenclature 8-10-6-6-6-10-0.2. 2
(iii) Name some cutting tool materials. 1
(iv) Mention the factors affecting tool life. 2
6. (i) Specify a centre lathe. 3
(ii) Why Face Plate and Angle Plates are used in Lathe Machine? 2
(iii) Name any three types of lathe operation showing their schematic representation. 3



3236244

7. (i) Draw different types of oxy-acetylene flame showing different zones. 3
(ii) When reverse polarity is used in arc welding? 2
(iii) What are the causes of using flux in arc welding? 2
(iv) Name any one field of application of TIG welding. 1
8. (i) What is match plate pattern? 2
(ii) Name any two types of pattern allowances. 2
(iii) Name the ingredients of moulding sand with their utility. 2
(iv) Write steps in investment casting. 2
9. (i) Discuss any one type of taper turning operation with figure. 3
(ii) Write down field of application of die and centrifugal casting. 2
(iii) Write down basic differences of brazing and soldering. 2
(iv) On what kind of material spot welding can be done? 1
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JANUARY 2024

MANUFACTURING PROCESSES - I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of 'Group: A' must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from 'Group: B'.

GROUP: A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): **2x10=20**

- i. The metal is subjected to mechanical working for –
a) Refining grain size b) Changing original block into desired shape c) Controlling the direction of flow lines d) All of these
- ii. The hot working of metals is carried out –
a) At the recrystallization temperature b) Below the recrystallization temperature c) Above the recrystallization temperature d) At any temperature
- iii. Structural sections such as rails, angles, I-beams are made by –
a) Hot rolling b) Hot drawing c) Hot piercing d) Hot extrusion
- iv. In indirect extrusion, which type ram is used?
a) Solid ram b) Hollow ram c) Curved ram d) None of these
- v. In a four high rolling mill, there are four rolls out of which –
a) One is working roll and three are backing up rolls b) Two are working rolls and two are backing up rolls c) Three are working rolls and one is backing up roll d) All of the four are working rolls
- vi. In oblique cutting of metals, the cutting edge of the tool is –
a) Perpendicular to the work piece b) Perpendicular to the direction of tool travel c) Parallel to the direction of tool travel d) Inclined at an angle less than 90° to the direction of tool travel
- vii. Among the following, which one is considered as production purpose Lathe?
a) Center Lathe b) Gap bed Lathe c) Engine Lathe d) Capstan and Turret Lathe
- viii. Which of the following is related with offsetting the tail stock?
a) Turning b) Facing c) Taper turning d) Knurling
- ix. In order to hold an irregular cross-sectional job in a Lathe, which type of chuck is suggested?
a) Three jaw self-centered chuck b) Four jaw independent chuck c) Collet chuck d) Magnetic chuck
- x. The time elapse between two successive grinding of a cutting tool is known as –
a) Machining time b) Tool life c) Set-up time d) None of these
- xi. The main functions of cutting fluids are –
a) Lubrication b) Cooling c) Lubrication and cooling d) None of these
- xii. Which form of metal is used in casting process?
a) Solid b) Liquid c) Gas d) Plastic
- xiii. In which casting process master pattern and wax pattern are used?
a) Continuous casting b) Centrifugal casting c) Investment casting d) Shell casting
- xiv. Among the following, which one is fall under the category of pressure welding?
a) TIG welding b) Arc welding c) Resistance welding d) None of these
- xv. Among the following, which welding process does not require any electric power supply during its execution?
a) TIG welding b) MIG welding c) Thermit welding d) Resistance welding

GROUP: B

2. a) State the difference between hot working and cold working processes.
b) State the advantages and disadvantages of forging process.
3. a) With a suitable diagram explain the working of Four High Rolling Mill.
b) Describe with neat sketch the working principle of Direct Extrusion Process.

[4+4]

[4+4]

4. a) Define Tool Life of a cutting tool. How tool life of a cutting tool can be measured?
b) Explain the given tool signature of a single point cutting tool: $10^\circ, 8^\circ, 7^\circ, 6^\circ, 15^\circ, 30^\circ, 1/8$ (inch)

[3+5]

5. a) Explain the forces acting during oblique cutting operation by a single point cutting tool.
b) What are the desirable properties of cutting fluid?

[3+5]

6. a) What is taper? How taper can be represented?

b) A shaft 600 mm long is to be turned taper for a length of 225 mm. The amount of taper is 1:100. If the shaft diameter is 75 mm, determine the smaller diameter, half taper angle and set-over required to turn the taper.

[3+5]

7. With suitable diagram explain the process of investment casting.

[8]

8. a) State about the defects which may generate in casting process.

b) With suitable diagram explain the working principle of arc welding.

[3+5]

9. a) Explain the different types of welding flames used in gas welding.

b) State the difference between MIG welding and TIG welding.

[4+4]

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MANUFACTURING PROCESS-I*Time Allowed: 3 Hours**Full Marks: 60***Answer the following questions from Group-A, B & C as directed.****Group A**

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. An example of resistance welding is - a) TIG, b) MIG, c) Submerged arc welding, d) Spot welding.
- ii. Non-Consumable electrode used in- a) GMAW, b) GTAW, c) MMAW, d) Thermit.
- iii. A welding defect is - a) crack, b) crash, c) scratch, d) swell.
- iv. In which of the following processes, material is neither added nor removed but is deformed into desired shape?- (a) Surface finishing process, (b) Metal forming process, (c) Casting, (d) Machining.
- v. Mechanical property desired for successful forging is - a) Ductility, b) Malleability, c) Hardness, d) Toughness.
- vi. The most commonly used flame in gas welding is - a) Neutral b) Oxidizing c) Carburizing d) All of the above.
- vii. Structural section such as rails, angles, I beam are made by - a) Hot extrusion, b) Hot rolling, c) Hot drawing, d) Hot spinning.
- viii. Half nut is connected with - a) Milling machine, b) Locking device, c) Jigs & Fixture, d) Thread cutting on Lathe.
- ix. Long wires are made by following process - a) Extrusion, b) Rolling, c) Piercing, d) Drawing.
- x. A moving mandrel is used in - (a) wire drawing, (b) tube drawing, (c) metal cutting, (d) casting.
- xi. Which of the following flame is harmful to steel? - (a) Neutral flame, (b) Oxidizing flame, (c) Carburizing flame, (d) Both oxidizing flame and carburizing flame.
- xii. Which of the following materials is used as the pattern material in precision investment casting? a) Metal, b) Molten wax, c) Plastics, d) Wood.
- xiii. Blanking and punching operations can be performed simultaneously on – (a) compound die, (b) combination die, (c) progressive die, (d) simple die.
- xiv. A casting defect which results in general enlargement of a casting is known as- (a) Shift, (b) sand wash, (c) swell, (d) scab.
- xv. Core print is a part of - a) core, b) pattern, c) core box, d) chaplet.

2. Fill in the blanks (any ten):

1x10

- i. Extra metal which settles down in the gutter is known as _____.
- ii. The best shape for sprue in sand-casting is _____.
- iii. Voltage required for Arc welding is _____ volts.

- iv. Oxygen cylinder used in oxy-acetylene gas welding is usually painted in _____ colour.
- v. In sand moulding, the middle part of the flask is called _____.
- vi. A _____ is a tool used for enlarging a previously drilled hole.
- vii. In blanking operation, clearance is provided on _____.
- viii. Collets, angle plates and mandrel are _____ of a lathe.
- ix. A small depression in the parent metal during arc welding is called _____.
- x. Lathe is specified by _____ of the job over carriage.
- xi. Large size bolt heads are made by _____ forging.
- xii. Model of an object which is used for making mould in sand moulding is called _____.
- xiii. Collapsible paste tubes are produced by _____.
- xiv. In _____ process, a specified shape of metal from the side or edge of a metal sheet can be removed using suitable punch and die.
- xv. _____ is a mixture of finely divided aluminium and iron oxide.

3. Answer the following questions (any ten):

1x10

- i. What is a chaplet?
- ii. Why electrodes are coated in manual arc welding?
- iii. State the relationship between lead and pitch of a thread?
- iv. Sketch three types of gas welding flames.
- v. Which sheet metal operation is used to make coins?
- vi. What is used to reduce friction during machining operation?
- vii. Define orthogonal cutting.
- viii. Pressurised gating system is used for casting of which metals and alloys?
- ix. Write down Taylor tool life equation?
- x. What is the use of filler metal in welding?
- xi. What is core print?
- xii. Which manufacturing process is used for manufacturing of Crankshaft?
- xiii. Define tool life for a single point HSS cutting tool in lathe operation.
- xiv. Which welding is used for sheet metal?
- xv. In which furnace cast iron is melted?

Group B

4. Answer the following questions (any six):

2x6

- i. What is the difference between brazing and soldering?
- ii. State the purpose of using cutting fluids in machining operations.
- iii. Name different defects of rolling process.
- iv. Give examples of single point cutting tool & multi point cutting tool.
- v. State specification of a Centre lathe.
- vi. Write two advantages of submerged arc welding.
- vii. What is Carburising flame?
- viii. Differentiate between hot working process and cold working process.
- ix. What are the components of a carriage?
- x. What is swell?

Group C

5. Answer the following question (any one):

1x6

- i. How manufacturing processes are classified? Explain with example.
- ii. Explain hydrostatic extrusion process with a simple diagram.
- iii. Briefly explain the processes - a) cutting, b) lancing, c) embossing.

5+1

6

2+2+2

6. Answer the following question (any one): 1x6
- i. Write down the favourable conditions for continuous and discontinuous chip formation? 6
 - ii. Write short notes on – a) Facing, b) Parting-off, c) Knurling. 2+2+2
 - iii. What are the Desirable properties of cutting tool materials and cutting fluid? 3 + 3
7. Answer the following question (any one): 1x6
- i. Describe the processes TIG and MIG briefly. 3+3
 - ii. Describe submerged arc welding process with sketch. 6
 - iii. Name the various types of defects that occur in sand castings. State causes and remedies in brief. 2+4
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March 2021

MANUFACTURING PROCESSES – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any twenty): 1x20
- i) Fluxes are used while welding – (a) To increase the rate of welding, (b) To clean the joint, (c) To prevent oxidation of metal during welding, (d) All the above.
 - ii) Oxy-acetylene flame is used for welding of – (a) Monel, (b) Nickel, (c) Steel, (d) Bronze.
 - iii) The hidden defect of a weld can be detected by – (a) Tensile test, (b) Bend test, (c) Radiography, (d) All.
 - iv) Commonly used gases in TIG welding are – (a) Hydrogen & Oxygen, (b) Hydrogen & Argon, (c) Hydrogen & Helium, (d) Helium & Argon.
 - v) Consumable electrode is used in the following process – (a) TIG welding, (b) Gas welding, (c) Arc welding, (d) Resistance welding.
 - vi) Above _____ temperature, working of metals is named hot working.
 - vii) Drawing is done to _____ the diameter of wire.
 - viii) In 4-high rolling mill, bigger rolls are known as _____ rolls.
 - ix) Accurate taper and step tapers can be machined with _____ attachment.
 - x) Finishing and sizing a hole is called _____.
 - xi) Spot welding is one of the _____ welding processes.
 - xii) The revolving cutter of a milling machine is held on a _____.
 - xiii) In universal milling machine table can have the _____ arrangement about a vertical axis.
 - xiv) Mechanical property desired for successful forging is – (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness
 - xv) Seamless tubes in mass production are manufactured by – (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - xvi) In drop forging, forging is done by dropping – (a) The work piece at high velocity, (b) The hammer at high velocity, (c) The die with hammer with high velocity, (d) A weight on hammer to produce the requisite impact.
 - xvii) Blanking and punching operations can be performed simultaneously on – (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - xviii) A twist drill has _____ flutes.
 - xix) Pattern used in precision investment casting normally made of _____.

- xx) In resistance welding _____ is generated by the resistance offered to the flow of current.
- xxi) MIG welding uses _____ electrode.
- xxii) In neutral flame in Oxyacetylene gas welding, ratio of acetylene to oxygen is _____.

2. What is core print? What is core? Write the advantages of wood as pattern making material. 2+3+5
 3. How a Lathe machine can be specified? Write the functions of Lathe machine parts? 5+5
 4. Write the difference between up milling and down milling. Write the specifications of drilling machine. 5+5
 5. Write shorts on – (a) Facing, (b) Parting-off, (c) Knurling. 3+3+4
 6. Write the functions of different parts of a radial drilling machine. 10
 7. Describe hot chamber and cold chamber die-casting method. 5+5
 8. Write short notes on – (a) Sweep pattern, (b) Loose-piece pattern. 5+5
 9. Describe submerged arc welding process. 10
 10. Describe different types of milling cutter. 10
 11. Write short notes on different types of chips. 10
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MANUFACTURING PROCESSES-I**841**

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier. Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty):

1x20

- i. Which of the following processes is mainly used for making the connecting rod?
a) Casting b) Deep drawing c) Rolling d) Forging
- ii. The hot working of metals is carried out at _____.
a) The recrystallisation temperature b) Below the recrystallisation temperature c) Above the recrystallisation temperature d) Any temperature
- iii. Which of the following is a type of extrusion process?
a) Direct b) Indirect c) Impact d) All of the mentioned
- iv. Long wires are made by _____ process.
a) Forging b) Rolling c) Piercing d) Drawing
- v. A moving mandrel is used in _____.
a) Wire drawing b) Tube drawing c) Metal cutting d) Casting
- vi. A two high rolling mill consists of two rolls which rotate _____.
a) At the same speed and in the same direction b) At the same speed but opposite direction c) At different speeds and in the same direction d) At the different speeds and in the opposite direction
- vii. In press, which of the following mechanism / system may be used for applying power to ram?
a) Rack and pinion b) Pneumatic c) Hydraulic d) All of these
- viii. In which sheet metal operation, the punch out portion of the sheet metal is the useful part?
a) Blanking b) Piercing c) Coining d) Embossing
- ix. Which of the following part is used for holding the metal sheet during blanking operation?
a) Spherical steel ball b) Roller c) Pressure pad d) Magnet
- x. Which type of chip is formed during machining of soft and ductile material?
a) Continuous chip b) Discontinuous chip c) continuous chip with built-up edge d) None of these
- xi. How many forces are acting during orthogonal cutting action?
a) One b) Two c) Three d) Four
- xii. Which type of standard taper is being provided on the shank of Lathe centers?
a) Morse taper b) Jarno taper c) Brown and Sharpe taper d) None of these
- xiii. Pitch of the lead screw of a metric Lathe is _____.
a) 4 mm b) 6 mm c) 8 mm d) 10 mm
- xiv. The size of a Lathe is specified by the _____.
a) Length between centers b) Swing diameter over the bed c) Swing diameter over the carriage d) All of these
- xv. To perform the facing operation in a Lathe, which type of feed has to be given on cutting tool?
a) Longitudinal feed b) Cross feed c) Inclined feed d) None of these
- xvi. Which specific gear is required in change gear for cutting metric thread in an English Lathe?
a) Gear of 20 teeth b) Gear of 40 teeth c) Gear of 80 teeth d) Gear of 127 teeth

- xvii. During machining, cutting fluids are used for:
a) Cool the cutting tool b) Wash away the chips c) Cool the workpiece d) All of these
- xviii. In which Lathe operation, the effective diameter of the job get increased?
a) Turning b) Facing c) Grooving d) Knurling
- xix. Which allowance is incorporated in pattern making to compensate the shrinkage of metal during solidification?
a) Machining allowance b) Distortion allowance c) Draft allowance d) Shrinkage allowance
- xx. In order to produce hollow interior of the casting _____ are used.
a) Riser pins b) Cores c) Vent tubes d) None of these
- xxi. "Master pattern" and "lost pattern" are used in _____ casting process.
a) Continuous b) Centrifugal c) Investment d) None of these
- xxii. In gas welding, how many types of flames are used for welding?
a) One b) Two c) Three d) Four
- xxiii. In which gas welding flame three distinct cones are obtained?
a) Oxidizing flame b) Carburizing flame c) Neutral flame d) None of these
- xxiv. In TIG welding, tungsten acts as _____ electrode.
a) Consumable b) Non-consumable c) Filler d) None of these
- xxv. Which welding process requires filler material?
a) Metal arc welding b) MIG welding c) TIG welding d) Resistance welding

Group B

- 2.a. What is hot forging and cold forging?
b. With suitable diagram explain the open die forging process. 2+6
- 3.a. State the major advantages of forging process.
b. With suitable diagram explain the indirect extrusion process. 2+6
- 4.a. State the difference between orthogonal and oblique cutting.
b. State the desirable properties of cutting tool materials. 4+4
- 5.a. Define longitudinal, cross and inclined feed in respect of Lathe.
b. With suitable diagram show the nomenclature of a single point cutting tool. 3+5
- 6.a. How will you specify a Lathe?
b. State the difference between steady rest and follower rest. 4+4
- 7.a. Define odd thread and even thread.
b. Calculate the teeth number of gears and draw the gear arrangement of the change gear for cutting a screw having 7 mm pitch on a Lathe having lead screw of 4 threads per inch. 2+6
- 8.a. State the name of allowances generally being incorporated in pattern making.
b. With suitable diagram explain the centrifugal casting process. 2+6
- 9.a. State the advantages and disadvantages of welding process.
b. State the difference between gas welding and arc welding. 4+4
- 10.a. State the function of flux on a coated electrode used in arc welding.
b. With suitable diagram explain the principle of submerged arc welding. 3+5